FORMATIVE ASSESSMENT 2
LAB ACTIVITY

1. Aim

To find the focal length of a given convex lens by using the u-v method.

2. Apparatus

- Convex lens
- Lens stand (V-stand)
- White screen
- Meter scale
- Illuminated object (a candle or illuminated wire gauze)

3. Procedure

1. First, determine the approximate focal length of the convex lens by focusing a distant object (like a tree or a distant building) onto the screen. The distance between the lens and the sharp image on the screen is the approximate focal length.
2. Mount the convex lens on the lens stand and place it on a long table over a meter scale.
3. Place the illuminated object (candle) at a known distance (u) in front of the lens. Ensure this distance is greater than the approximate focal length.
4. Place the screen on the opposite side of the lens and move it back and forth along the scale.
5. Stop adjusting when a sharp, clear, and inverted image of the object is formed on the screen.
6. Record the distance between the optical center of the lens and the object as object distance (u), and the distance between the lens and the screen as image distance (v).

7. Repeat the experiment for at least four different object distances and tabulate the readings.

4. Observations

The observed values of object distance (u) and image distance (v) are recorded in a tabular format:

S.No.	Object Distance (u) cm	Image Distance (v) cm	Focal Length (f) cm
1			
2			
3			
4			

5. Calculations & Analysis

According to the sign convention, the object distance is taken as negative ($-u$) and the real image distance is taken as positive ($+v$).

The lens formula is given by:

$$1/f = 1/v - 1/u$$

Substituting the sign conventions, the magnitude of the focal length is calculated as:

$$f = (u \times v) / (u + v)$$

The average of the focal lengths calculated for different sets of readings gives the final accurate focal length of the lens.

6. Result

The focal length of the given convex lens, as determined by the ***u-v*** method, is found to be _____ **cm**.

7. Precautions

1. The principal axis of the convex lens must be perfectly horizontal and parallel to the measuring scale.
2. The object (flame/wire gauze), the optical center of the lens, and the screen should all be aligned at the same height in a straight line.
3. Proper sign conventions must be applied meticulously while calculating the focal length to avoid errors.

